



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

CHEMISTRY

0620/62

Paper 6 Alternative to Practical

May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

IGCSE™ is a registered trademark.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **6** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	(gas) syringe	1
1(b)(i)	volume of gas / volume of carbon dioxide	1
1(b)(ii)	reaction finished / no more gas given off	1
	calcium carbonate used up	1
1(c)	sketch less steep at beginning	1
	to same level / volume / final amount of gas	1
1(d)	limewater / calcium hydroxide solution	1
	milky / cloudy / white ppt.	1

Question	Answer	Marks
2(a)	initial temperatures all 21 AND final temperatures 18, 17, 15	1
	temperature changes –3, –4, –6	1
2(b)	initial temperatures 22, 22, 21, 22 AND final temperatures 26, 27, 29, 33	1
	temperature changes +4, +5, +8, +11	1
2(c)	all points plotted correctly ($\pm$ half a small square)	2
	best-fit straight-line graphs	1
	labels D (upper) and C (lower) or (expt.) 2 and 1	1

Question	Answer	Marks
2(d)	value from graph, -8°C	1
	extrapolation	1
2(e)	exothermic	1
2(f)	room temperature / 21°C / 22°C	1
	heat lost to surroundings	1
2(g)	half as much	2
2(h)	change in apparatus or method e.g. use a pipette / burette or use insulation / lid	1
	explanation e.g. as more accurate / precise, than a <u>measuring cylinder</u> / reduce heat losses	1
2(i)	repeat experiments	1
	compare / average	1

Question	Answer	Marks
3(a)	(pale) green (solid / crystals)	1
3(b)	no change / no reaction / no precipitate / no observation	1
3(c)	white precipitate	1
3(d)	green	1
	precipitate	1
3(e)	green precipitate	1
3(f)	calcium	1

Question	Answer	Marks
4	any 6 from: hydrochloric acid in burette / measuring cylinder (solutions can be reversed) measured volume of barium hydroxide solution (solutions can be reversed) in named container e.g. beaker / (conical) flask (named) indicator (ignore Universal Indicator) OR pH meter acid added gradually / slowly / dropwise / dripped until colour changes / endpoint / neutral / pH 7 note volume added / initial and final volumes calculation (using volumes and concentration of the acid)	max 6